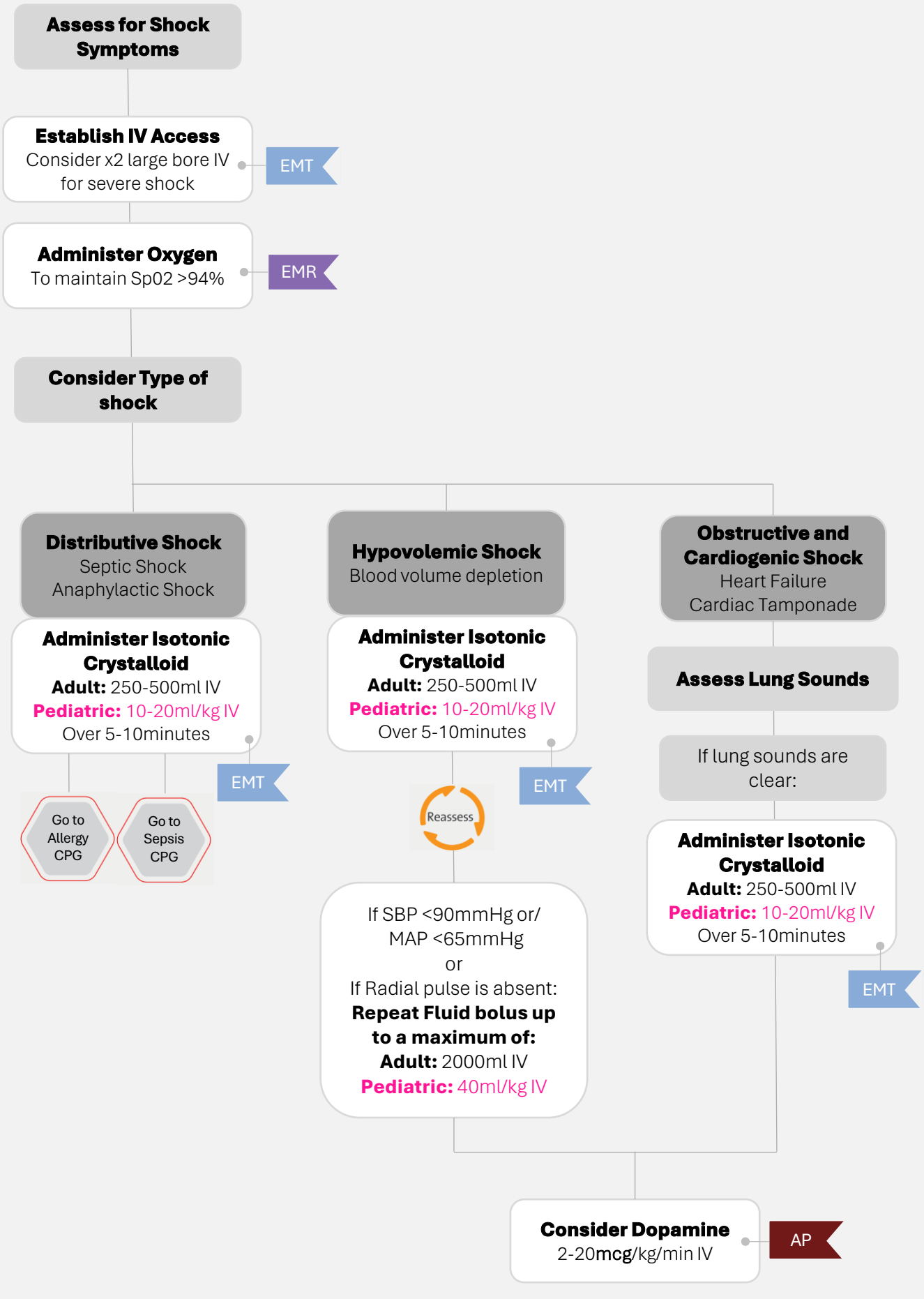
Non-Traumatic Shock



Patient Care Goals

1. Identification and management of life-threatening non-traumatic shock (e.g., hypovolemic, cardiogenic, distributive, or obstructive).
2. Improve the patient's hemodynamic status through administration of fluids, vasopressors, and supportive care.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms of shock which is not related to trauma, including:
  - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics).
  - Tachycardia (HR >100 in adults, or age-specific for pediatrics).
  - Poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
  - Altered mental status, weakness, or respiratory distress.
- Suspected underlying causes such as sepsis, anaphylaxis, cardiogenic shock, or internal bleeding (e.g., gastrointestinal or ectopic pregnancy).

Exclusion Criteria

- Shock clearly attributable to trauma.

Key Patient Assessment Points

1. **Shock Type Identification**
   - Assess for clues to differentiate shock types:
     - Hypovolemic: History of fluid loss (e.g., vomiting, diarrhea, or occult bleeding).
     - Cardiogenic: Chest pain, dyspnea, or history of heart disease.
     - Distributive: Signs of sepsis (e.g., fever, infection) or anaphylaxis (e.g., urticaria, wheezing).
     - Obstructive: Symptoms suggestive of pulmonary embolism or pericardial tamponade (e.g., jugular vein distension, muffled heart sounds).
2. **Perfusion Status**
   - Evaluate skin temperature, color, and moisture (e.g., cool, pale, or mottled skin).
   - Assess for signs of organ dysfunction (e.g., altered mental status, decreased urine output).
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary surveys as per Universal Care CPG.
   - Consider Oxygen
     - Titrate to maintain SpO2 ≥94% if hypoxia or respiratory distress is present.
2. **Consider Red Flags**
   - Persistent hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics) despite fluid resuscitation.
   - Severe altered mental status or unresponsiveness.
   - Hypoxia (SpO2 <90%) or respiratory failure.
   - Cardiogenic or Obstructive shock.
3. **Evaluate for Signs of Shock or Instability**
   - Hypotension, tachycardia, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
4. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
5. **Establish IV Access**
   - Indicated for hypotension, suspected shock, or anticipated medication/fluid administration.
   - Consider intraosseous (IO) access if IV access cannot be obtained.
6. **Consider Condition-Specific Treatment**
   - For Septic and Anaphylactic Shock, refer to appropriate CPG for additional treatment considerations and additional shock management.

7. **Fluid Resuscitation**
   - Fluid Administration Targets:
     - Fluid bolus' should be administered to target:
       - Adults: MAP (mean arterial pressure) = 65mmHg
       - Pediatrics: Systolic blood pressure of at least fifth percentile for age, strong distal pulses, warm skin perfusion, capillary refill less than 2 seconds and improving mental status.
   - Administer Normal Saline or Ringers Lactate:
     - Adults: 250-500 mL bolus IV, repeat as needed (up to 2000ml)
     - Pediatrics: 20 mL/kg bolus IV, repeat as needed (max 40 mL/kg).
   - Use caution in obstructive/cardiogenic shock; administer smaller boluses.
8. **Consider Vasopressors**
   - Indications:
     - Fluid refractory hypotension after fluid resuscitation (once fluid maximums have been reached).
     - Hypotension with Pulmonary Edema (e.g. Obstructive/Cardiogenic shock).
   - Dopamine
     - Adults: 5-20 mcg/kg/min IV infusion.
     - Pediatrics: 5-20 mcg/kg/min IV infusion.
9. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Elderly Patients
  - Shock may present atypically (e.g., confusion, minimal tachycardia due to beta-blocker use).
  - Use caution with fluid boluses in patients with heart failure; consult Tele EMS for guidance.
- Fluid Resuscitation:
  - Avoid over-resuscitation in cardiogenic or obstructive shock; monitor for signs of pulmonary edema (e.g., crackles, dyspnea).

Clinical Insights & Additional Guidance

- Common Causes:
  - Hypovolemic: Dehydration, gastrointestinal bleeding, or ruptured ectopic pregnancy.
  - Cardiogenic: Myocardial infarction, heart failure, or arrhythmias.
  - Distributive: Sepsis, anaphylaxis, or neurogenic shock (e.g., spinal cord injury).
  - Obstructive: Pulmonary embolism, pericardial tamponade, or tension pneumothorax.
- Possible Alternative Causes:
  - Traumatic shock, adrenal insufficiency, hypoglycemia, or drug overdose (e.g., antihypertensives).

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Primary complaint or symptoms such as dizziness, syncope, hypotension, or chest pain.
- Injuries / Information: Relevant history including recent illness, known allergies, chronic diseases, and medication use.
- Signs: Vital signs (BP, HR, RR, SpO_2), mental status, skin condition (pale, cool, clammy), capillary refill, GCS or AVPU.
- Treatment:
  - Oxygen therapy: rate, method, response.
  - IV fluid administration: type, volume, and response.
  - Vasopressors if used: name, dose, and response.
- Allergies: Drug or substance allergies relevant to medications administered.
- Medications: Current or recent medications including antihypertensives, diuretics, or vasodilators.

- Background: Comorbid conditions such as diabetes, heart failure, or recent infections.
- Other: 12-lead ECG findings, hospital notification, scene observations.

Key Performance Indicators

1. Administer IV fluid to hypotensive patients.

References

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